GATE ME 90 Days Challenge

Day 27/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Engineering Mathematics	Vector Calculus	80 days to GATE

How to use this set: 15 questions (6 MCQ, 9 NAT) on gradient, directional derivatives and normals, divergence and curl, irrotational and conservative fields, line integrals and work, and the Green, Gauss (divergence) and Stokes theorems. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

The magnitude of the gradient of $\varphi = x^2y + yz^3$ at the point (1, 2, 1) (round off to two decimal places) is _____

Q2 NAT

The directional derivative of $\varphi = x^2 + 2y^2 + 3z^2$ at the point (1, 1, 1) in the direction of the vector $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ (round off to two decimal places) is _____

Q3 MCQ

The unit normal to the surface $x^2 + y^2 - z = 1$ at the point (1, 1, 1) is

(A) $(\mathbf{i} + \mathbf{j} - \mathbf{k})/\sqrt{3}$
(B) $(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})/3$
(C) $(2\mathbf{i} + 2\mathbf{j} + \mathbf{k})/3$
(D) $(\mathbf{i} + \mathbf{j})/\sqrt{2}$

Q4 NAT

The divergence of $\mathbf{F} = x^2y\,\mathbf{i} + y^2z\,\mathbf{j} + z^2x\,\mathbf{k}$ at the point (1, 2, 3) is _____

Q5 MCQ

The curl of the vector field $\mathbf{F} = y\,\mathbf{i} - x\,\mathbf{j}$ is

(A) $2\mathbf{k}$
(B) $\mathbf{0}$
(C) $-2\mathbf{k}$
(D) $\mathbf{i} - \mathbf{j}$

Q6 MCQ

For twice continuously differentiable φ and $\mathbf{F}$, which of the following is **NOT** identically zero?

(A) $\nabla\times(\nabla\varphi)$
(B) $\nabla\cdot(\nabla\times\mathbf{F})$
(C) $\nabla\cdot[\nabla\times(\varphi\mathbf{F})]$
(D) $\nabla\cdot(\nabla\varphi)$

Q7 NAT

The vector field $\mathbf{F} = (x + 2y + az)\,\mathbf{i} + (bx - 3y - z)\,\mathbf{j} + (4x + cy + 2z)\,\mathbf{k}$ is irrotational. The value of $a + b + c$ is _____

Q8 NAT

The work done by the force $\mathbf{F} = 3x^2\,\mathbf{i} + (2xz - y)\,\mathbf{j} + z\,\mathbf{k}$ in moving a particle along the straight line from (0, 0, 0) to (2, 1, 3) is _____

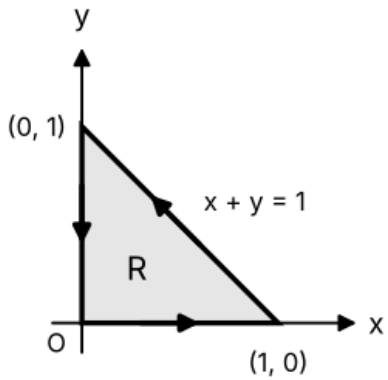
Q9 MCQ

The force $\mathbf{F} = (2xy + z^3)\,\mathbf{i} + x^2\,\mathbf{j} + 3xz^2\,\mathbf{k}$ moves a particle from (1, −2, 1) to (3, 1, 4). The work done is

(A) 202
(B) 200
(C) 198
(D) 204

Q10 NAT

Using Green's theorem, the value of $\oint_C [(3x - 8y^2)\,dx + (4y - 6xy)\,dy]$, where C is the boundary of the triangle bounded by $x = 0$, $y = 0$ and $x + y = 1$ traversed anticlockwise (figure), is _____ (round off to two decimal places)



Q11 NAT

The outward flux of $\mathbf{F} = x^3\,\mathbf{i} + y^3\,\mathbf{j} + z^3\,\mathbf{k}$ through the sphere $x^2 + y^2 + z^2 = 1$ (round off to two decimal places) is _____

Q12 MCQ

For $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$, the circulation $\oint \mathbf{F}\cdot d\mathbf{r}$ around the circle $x^2 + y^2 = 4$ in the plane $z = 0$, traversed anticlockwise, is

(A) 8π
(B) 4π
(C) 16π
(D) 0

Q13 NAT

Using the line integral $A = \tfrac{1}{2}\oint(x\,dy - y\,dx)$, the area enclosed by the ellipse $x^2/9 + y^2/4 = 1$ (round off to two decimal places) is _____

Q14 NAT

The outward flux of $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ through the surface of the cube $0 \le x, y, z \le 2$ is _____

Q15 MCQ

With $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = |\mathbf{r}| \neq 0$, $\nabla^2(r^n) = n(n+1)r^{n-2}$. Among the options, r^n is harmonic ($\nabla^2 r^n = 0$) for

(A) $n = -1$
(B) $n = 1$
(C) $n = 2$
(D) $n = -2$

Answer Key

Q	Type	Answer
Q1	NAT	7.48 (accept 7.44 to 7.52)
Q2	NAT	7.33 (accept 7.30 to 7.37)
Q3	MCQ	(B) $(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})/3$
Q4	NAT	22
Q5	MCQ	(C) $-2\mathbf{k}$
Q6	MCQ	(D) $\nabla\cdot(\nabla\varphi)$
Q7	NAT	5
Q8	NAT	16
Q9	MCQ	(A) 202
Q10	NAT	1.67 (accept 1.65 to 1.69)
Q11	NAT	7.54 (accept 7.50 to 7.58)
Q12	MCQ	(A) 8π
Q13	NAT	18.85 (accept 18.75 to 18.95)
Q14	NAT	24
Q15	MCQ	(A) $n = -1$

Detailed Solutions

Q1 Answer: 7.48 (accept 7.44 to 7.52)

Concept: $\nabla\varphi = (\partial\varphi/\partial x, \partial\varphi/\partial y, \partial\varphi/\partial z)$; its magnitude is the maximum rate of change of φ.

$\nabla\varphi = (2xy, x^2 + z^3, 3yz^2) = (4, 2, 6)$ at $(1, 2, 1)$

$|\nabla\varphi| = \sqrt{(16 + 4 + 36)} = \sqrt{56} = \mathbf{7.48}$

Exam tip: $\nabla\varphi$ points along the normal to the level surface φ = constant through the point.

Q2 Answer: 7.33 (accept 7.30 to 7.37)

Concept: Directional derivative = $\nabla\varphi \cdot \hat{\mathbf{a}}$, where $\hat{\mathbf{a}}$ is the **unit** vector of the given direction.

$\nabla\varphi = (2x, 4y, 6z) = (2, 4, 6)$; $\hat{\mathbf{a}} = (1, 2, 2)/3$

$D = (2 + 8 + 12)/3 = 22/3 = \mathbf{7.33}$

Exam tip: Forgetting to normalise the direction is the most common mistake: $(2, 4, 6)\cdot(1, 2, 2) = 22$ is wrong by a factor of 3.

Q3 Answer: (B) $(2\mathbf{i} + 2\mathbf{j} - \mathbf{k})/3$

Concept: For a surface $f(x, y, z) = c$, the normal is ∇f.

$f = x^2 + y^2 - z$: $\nabla f = (2x, 2y, -1) = (2, 2, -1)$, $|\nabla f| = 3$

Unit normal = $\mathbf{(2i + 2j - k)/3}$ (or its negative)

Exam tip: The angle between two surfaces at a common point is the angle between their gradient vectors.

Q4 **Answer: 22**

Concept: $\nabla\cdot\mathbf{F} = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$ (a scalar).

$\nabla\cdot\mathbf{F} = 2xy + 2yz + 2zx = 2(2) + 2(6) + 2(3) = \mathbf{22}$

Exam tip: Divergence measures the net outflow per unit volume; a field with zero divergence everywhere is solenoidal (incompressible flow).

Q5 **Answer: (C) −2k**

Concept: $\nabla\times\mathbf{F} = (\partial F_z/\partial y - \partial F_y/\partial z)\,\mathbf{i} + (\partial F_x/\partial z - \partial F_z/\partial x)\,\mathbf{j} + (\partial F_y/\partial x - \partial F_x/\partial y)\,\mathbf{k}$.

Only the **k** component survives: $\partial(-x)/\partial x - \partial(y)/\partial y = -1 - 1 = \mathbf{-2}$

Exam tip: This field is a clockwise rigid rotation with angular velocity 1 rad/s; curl = 2 × angular velocity, here $-2\mathbf{k}$.

Q6 **Answer: (D) ∇·(∇φ)**

Concept: Two identities hold for all smooth fields: curl grad φ = **0** and div curl **F** = 0 (mixed partial derivatives cancel).

$\nabla\cdot(\nabla\varphi) = \nabla^2\varphi$, the Laplacian, which is zero only for harmonic functions (e.g. $\varphi = x^2$ gives $\nabla^2\varphi = 2$).

Exam tip: Option (C) is zero too: it is div curl of the vector field $\varphi\mathbf{F}$.

Q7 **Answer: 5**

Concept: Irrotational means $\nabla\times\mathbf{F} = \mathbf{0}$: all three components of the curl vanish.

i: $\partial F_z/\partial y - \partial F_y/\partial z = c + 1 = 0 \rightarrow \mathbf{c = -1}$

j: $\partial F_x/\partial z - \partial F_z/\partial x = a - 4 = 0 \rightarrow \mathbf{a = 4}$

k: $\partial F_y/\partial x - \partial F_x/\partial y = b - 2 = 0 \rightarrow \mathbf{b = 2}$; $a + b + c = \mathbf{5}$

Exam tip: An irrotational field is conservative ($\mathbf{F} = \nabla\varphi$) on a simply connected region.

Q8 **Answer: 16**

Concept: $W = \int\mathbf{F}\cdot d\mathbf{r}$; parametrise the path and integrate in one variable.

Line: $x = 2t$, $y = t$, $z = 3t$ $(0 \le t \le 1)$, $d\mathbf{r} = (2, 1, 3)\,dt$

$\mathbf{F}\cdot d\mathbf{r} = [3(4t^2)(2) + (12t^2 - t)(1) + 3t(3)]\,dt = (36t^2 + 8t)\,dt$

$W = 12 + 4 = \mathbf{16}$

Exam tip: This field is not conservative ($\partial F_y/\partial x = 2z \neq \partial F_x/\partial y = 0$), so the work depends on the path.

Q9 **Answer: (A) 202**

Concept: $\nabla\times\mathbf{F} = \mathbf{0}$ here, so $\mathbf{F} = \nabla\varphi$ and $W = \varphi(\text{end}) - \varphi(\text{start})$, whatever the path.

By inspection $\varphi = x^2y + xz^3$ (check: $\partial\varphi/\partial x = 2xy + z^3$, $\partial\varphi/\partial y = x^2$, $\partial\varphi/\partial z = 3xz^2$)

$W = \varphi(3, 1, 4) - \varphi(1, -2, 1) = (9 + 192) - (-2 + 1) = \mathbf{202}$

Exam tip: Always test the curl first; a zero curl turns a long line integral into a one-line answer.

Q10 **Answer: 1.67 (accept 1.65 to 1.69)**

Concept: Green's theorem: $\oint(M\,dx + N\,dy) = \iint_R(\partial N/\partial x - \partial M/\partial y)\,dA$ for anticlockwise C.

$\partial N/\partial x - \partial M/\partial y = -6y - (-16y) = \mathbf{10y}$

$\iint 10y\,dA = 10\int_0^1\int_0^{1-x} y\,dy\,dx = 10 \times 1/6 = \mathbf{5/3 = 1.67}$

Exam tip: Direct evaluation along the three sides gives the same 5/3, a good check if time permits.

Q11 **Answer: 7.54 (accept 7.50 to 7.58)**

Concept: Gauss divergence theorem: $\oiint \mathbf{F}\cdot\mathbf{n}\,dS = \iiint \nabla\cdot\mathbf{F}\,dV$.

$\nabla\cdot\mathbf{F} = 3(x^2 + y^2 + z^2) = 3r^2$

$\iiint 3r^2\,dV = \int_0^1 3r^2 \times 4\pi r^2\,dr = 12\pi/5 = \mathbf{7.54}$

Exam tip: Use spherical coordinates ($dV = 4\pi r^2\,dr$ for a radial integrand) whenever the divergence depends only on r.

Q12 **Answer: (A) 8π**

Concept: Stokes' theorem: $\oint\mathbf{F}\cdot d\mathbf{r} = \iint(\nabla\times\mathbf{F})\cdot\mathbf{n}\,dS$.

$\nabla\times\mathbf{F} = 2\mathbf{k}$, $\mathbf{n} = \mathbf{k}$, area $= \pi(2)^2 = 4\pi$

Circulation $= 2 \times 4\pi = \mathbf{8\pi}$

Exam tip: Check directly: on r = 2, $\mathbf{F}\cdot d\mathbf{r} = r^2\,d\theta = 4\,d\theta$, giving 8π over one turn.

Q13 **Answer: 18.85 (accept 18.75 to 18.95)**

Concept: Green's theorem with $M = -y/2$, $N = x/2$ gives $\partial N/\partial x - \partial M/\partial y = 1$, so the integral equals the area.

$x = 3\cos\theta$, $y = 2\sin\theta$: $x\,dy - y\,dx = (6\cos^2\theta + 6\sin^2\theta)\,d\theta = 6\,d\theta$

$A = \tfrac{1}{2} \times 6 \times 2\pi = 6\pi = \mathbf{18.85}$ $(= \pi ab)$

Exam tip: $\oint x\,dy$ and $-\oint y\,dx$ each also give the area on their own.

Q14 **Answer: 24**

Concept: Divergence theorem: flux $= \iiint\nabla\cdot\mathbf{F}\,dV$.

$\nabla\cdot\mathbf{F} = 1 + 1 + 1 = 3$; volume $= 2^3 = 8$

Flux $= 3 \times 8 = \mathbf{24}$

Exam tip: For the position vector **r**, the flux through any closed surface is 3 × the enclosed volume.

Q15 **Answer: (A) n = −1**

Concept: $\nabla^2(r^n) = n(n + 1)\,r^{n-2}$ vanishes for $n = 0$ or $n = -1$.

So $1/r$ is harmonic away from the origin: it is the potential of a point source (or a point mass/charge).

Exam tip: Related results: $\nabla r^n = n\,r^{n-2}\mathbf{r}$, $\nabla\cdot\mathbf{r} = 3$ and $\nabla\times\mathbf{r} = \mathbf{0}$.

What next?

Day 27 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 28/90: Fluid Mechanics, Fluid Properties & Statics. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.